

XV6 Problem Set 1 and First Assignment

Problem Set 1

```
6a5cc0ff.67dcf76 problemset1 -> problemset1
short@phoria-virtual-machine: ~/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_find user/_sixfive user/_sleep user/_cat user/_echo user/_forktest user/_grep user/_lnt user/_kill user/_ln user/_ls user/_mkdir user/_rm
user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_mendump
meta: 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1005 blocks have been allocated
ballocc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -n 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,d
rive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ sixfive README
$
1810
$
1810
```

Figure 1: Sixfive Manual

```
short@phoria-virtual-machine: ~/xv6-labs-2025$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_find user/_sixfive user/_sleep user/_cat user/_echo user/_forktest user/_grep user/_lnt user/_kill user/_ln user/_ls user/_mkdir user/_rm
user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_mendump
meta: 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 1005 blocks have been allocated
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qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -n 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,d
rive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
$ find . b
./b
./a/b
./a/aa/b
```

Figure 2: Find Manual

```
xv6 kernel is booting

hart 2 starting
hart 1 starting
init: starting sh
$ memdump
Example 1:
61810
2025
Example 2:
a string
Example 3:
another
Example 4:
8D0
1819438967
100
z
xyzyz
Example 5:
hello
w
o
r
l
d
$ echo abcdefgh12345678 | memdump p
64636261
$ echo abcdefgh12345678 | memdump s
abcdefgh12345678

$ echo abcdefgh12345678 | memdump ii
1684234849
1751606885
$ echo deadc0de | memdump hhcccc
25956
25697
c
0
d
e
a
```

Figure 3: Memdump Manual

```
sharlagheria@virtual-machine: ~/xv6-labs-2021$ make qemu
mv -f fs.img fs.img.bk
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_find user/_sixfive user/_sleep user/_cat user/_echo user/_forktest user/_grep user/_tnt user/_kill user/_ln user/_ls user/_mkdir user/_rm
user/_sh user/_stressfs user/_usertests user/_grind user/_gc user/_zombie user/_logstress user/_forphan user/_dorphan user/_memdump
meta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2080
balloc: first 1065 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,d
rive=x0,bus=virtio-mmio-bus.0

xv6 kernel is booting

hart 1 starting
hart 2 starting
init: starting sh
$ echo > b
$ mkdir a
$ echo > a/b
$ mkdir a/aa
$ echo > a/aa/b
$ find . b
./b
./a/b
./a/aa/b
$ find . b -exec echo
./b
./a/b
./a/aa/b
$
```

Figure 4: Exec Manual

```
ghorix@ghorix-virtual-machine: ~/xv6-labs-2025

== Test sleep, makes syscall ==
$ make qemu-gdb
sleep, makes syscall: OK (1.1s)
== Test sixfive_test ==
$ make qemu-gdb
sixfive_test: OK (0.9s)
== Test sixfive_readme ==
$ make qemu-gdb
sixfive_readme: OK (0.9s)
== Test sixfive_all ==
$ make qemu-gdb
sixfive_all: OK (1.1s)
== Test mendum, examples ==
$ make qemu-gdb
mendum, examples: OK (1.0s)
== Test mendum, format ti, S, p ==
$ make qemu-gdb
mendum, format ti, S, p: OK (1.0s)
== Test find, in current directory ==
$ make qemu-gdb
find, in current directory: OK (1.1s)
== Test find, in sub-directory ==
$ make qemu-gdb
find, in sub-directory: OK (1.0s)
== Test find, recursive ==
$ make qemu-gdb
find, recursive: OK (1.3s)
== Test exec ==
$ make qemu-gdb
exec: OK (0.8s)
== Test exec, multiple args ==
$ make qemu-gdb
exec, multiple args: OK (1.0s)
== Test exec, recursive find ==
$ make qemu-gdb
exec, recursive find: OK (1.4s)
== Test time ==
time: FAIL
    Cannot read time.txt
Score: 130/131
make: *** [Makefile:362: grade] Error 1
ghorix@ghorix-virtual-machine: ~/xv6-labs-2025$
```

Figure 5: Problem Set 1 Score

First Assignment

```
hart 1 starting
hart 2 starting
t0tt: starting sh
$ history
1 history
$ sh < README
too many args
syntax
too many args
exec ACKNOWLEDGMENTS failed
too many args
syntax
leftovers: )); See also https://pdos.csail.mit.edu/6.818/, which provides
syntax
exec pointers failed
syntax
leftovers: );, Cliff Frey (MP), Xiao Yu (MP), Nikolai Zeldovich, and Austin
syntax
exec Clements, failed
too many args
exec Abhinavpatel00, failed
exec Boyd-Wickizer, failed
too many args
exec echlweiner, failed
exec Nathaniel, failed
exec Hada, failed
exec Juley, failed
exec Wolfgang, failed
exec Kolontsov, failed
exec Handong, failed
exec nes900903, failed
exec OptIntititCside, failed
exec chuo, failed
exec Sehayek, failed
too many args
too many args
too many args
too many args
exec ERROR failed
too many args
leftovers: ... The main purpose of xv6 is as a teaching
```

Figure 6: Shell Testing

```
leftovers: . . The main purpose of xv6 is as a teaching
syntax
too many args
exec simplifications failed
exec BUILDING failed
exec You failed
exec https://github.com/riscv/riscv-gnu-toolchain, failed
exec riscv64-software failed
exec search failed
$ echo hi
hi
$ !!
echo hi
hi
$ QEMU: Terminated
ghorix@ghorix-virtual-machine:~/xv6-labs-2023$
```

Figure 7: Shell Testing 2

```
xv6 kernel is booting
hart 1 starting
hart 2 starting
init: starting sh
$ echo > b
$ echo > b
exec echo failed
$ echo > b
$ mkdir a; echo > a/b
$ mkdir a/aa; echo > a/aa/b
$ find . b
./b
./a/b
./a/aa/b
$ find . '*.bs' -re
$ find . '*.bs' -re
$ find . '*.BS' -re -exec echo
$ find . '*.BS' -re
./b
./a/b
./a/aa/b
$ find . '*.BS' -re -exec echo
exec find failed
$
```

Figure 8: Find Regex

```
ghorix@ghorix-virtual-machine:~/xv6-labs-2023$ git checkout -b firstAssignment
Switched to a new branch 'firstAssignment'
ghorix@ghorix-virtual-machine:~/xv6-labs-2023$ nano user/uptime.c
ghorix@ghorix-virtual-machine:~/xv6-labs-2023$ nano Makefile
ghorix@ghorix-virtual-machine:~/xv6-labs-2023$ make qemu
mv -f fs.img fs.img.bk
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcpu=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-math -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -fno-builtin-stack-protector -fno-pie -no-pie -c -o user/uptime.o user/uptime.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/uptime.user/ulib.o user/usys.o user/print.o user/unalloc.o
riscv64-linux-gnu-objdump -S user/uptime > user/uptime.asm
riscv64-linux-gnu-objdump -t user/uptime | sed 's|SYMBOL TABLE|d; s/ / /; /$$/d' > user/uptime.sym
riscv64-linux-gnu-gcc -Wall -Werror -O -fno-omit-frame-pointer -gdwarf-2 -DSOL_UTIL -DLAB_UTIL -MD -mcpu=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -fno-math -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -fno-builtin-stack-protector -fno-pie -no-pie -c -o user/mendump.o user/mendump.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/mendump.user/ulib.o user/usys.o user/print.o user/unalloc.o
riscv64-linux-gnu-objdump -S user/mendump > user/mendump.asm
riscv64-linux-gnu-objdump -t user/mendump | sed 's|SYMBOL TABLE|d; s/ / /; /$$/d' > user/mendump.sym
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/uptime.user/cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_userstest user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_mendump
qemu: 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
ballocc: first 983 blocks have been allocated
ballocc: write bitmap block at sector 48
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,driver=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ uptime
09
$
```

Figure 9: Uptime